Temporal-Spatial Decoupled Self-Supervised Multi-Task Learning for Video Anomaly Detection and Localization in Intelligent Transportation Surveillance Systems

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***Abstract*— Video anomaly detection (VAD) has become a critical research focus in the realms of computer vision and intelligent transportation systems, particularly with the rise of generative AI and advancements in self-supervised learning. These developments open up new frontiers in understanding complex scenarios within intelligent transportation systems. Since previous methods, confined to a single paradigm, inadequately captured intricate feature information, which led to a bottleneck that constrained the performance; individuals form unbiased and logical evaluations of matters through an in-depth review of a multitude of factors and rationalizations. Decoupled multi-task learning not only overcomes the limitations of single-task paradigms but also meets the requirements for comprehensiveness and objectivity in distinguishing things. Therefore, we propose a self-supervised multi-task learning network with object-level spatiotemporal decoupling in this paper. We decouple the video anomaly detection task in both spatial and temporal dimensions to construct multiple proxy tasks, optimizing overall performance through joint learning. Specifically, we first utilize a pre-trained object detector to construct spatiotemporal context cubes. Then, on the decoupled dimensions, we design several related proxy tasks, adjusting the context cube structure to accommodate each proxy task, and providing reliable information for anomaly detection through joint training. In the end, our method achieved performance that surpasses existing methods on three traffic road datasets, namely UCSD Ped2, Avenue, and ShanghaiTech, with AUCs of 99.6%, 95.6%, and 87.7%, respectively. It demonstrates the potential of generative AI-powered anomaly detection systems in advancing intelligent transportation applications.**

***Index Terms*— Video anomaly detection, multi-task learning, decoupling, generative AI, self-supervised Learning.**

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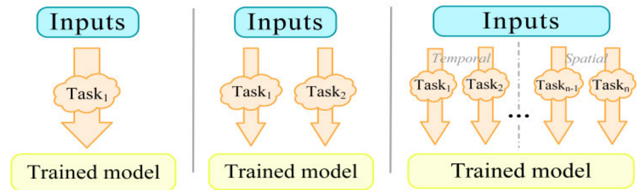
I. INTRODUCTION

WITH the rapid development of intelligent transportation systems, video surveillance, as a critical component of traffic management, serving as one of the core technologies for real-time traffic monitoring and anomaly behavior detection. In particular, pedestrian anomaly behavior detection in road scenarios is not only vital for traffic safety but also a key task in enhancing the intelligence level of intelligent transportation systems [1], [2]. In recent years, driven by an exponential surge in surveillance video, video anomaly detection continues to maintain its indispensability within the fields of computer vision and pattern recognition. Video anomaly detection, as a crucial component of intelligent video analysis, is widely applied in various urban management sectors, including intelligent transport system, autonomous driving, and infrastructure maintenance, within the construction of smart cities [3]. Additionally, in intelligent transportation systems [4], real-time identification and processing of abnormal events in traffic flow (such as traffic accidents, violations, emergencies, etc.) can significantly improve traffic safety and management efficiency, promote the intelligent and digital transformation of urban traffic management, and optimize traffic flow and resource allocation. Hence, augmenting the performance and efficiency of video anomaly detection holds paramount significance and value in realizing intelligent applications and propelling relevant research endeavors forward.

Up to the present, various anomaly detection methods have emerged under different supervisory settings, encompassing remarkable techniques in unsupervised [5], [6], [7], [8], [9], [10], [11], [12], weakly supervised [13], [14], and semi-supervised paradigms [15], [16]. In these regimes, early reconstruction or prediction methods simply utilized either motion or appearance information of objects for anomaly detection. However, Cai et al. [6] combined both to capture the consistency between motion and appearance. Although this combination has somewhat advanced the progress in addressing the consistency issue, the vanilla fusion of motion appearances may overlook the fact that the same type of abnormality could exhibit different patterns. In order to further explore higher-level visual features and investigate the complex relationship between object motion and appearance. Yang et al. [11] proposed a new paradigm for video anomaly detection based on keyframe reconstruction, grounded in video encoding theory. Although this method excels in maintaining video continuity and consistency, its effectiveness heavily depends on the representativeness of the selected keyframes. In the real world, abnormal events are relatively rare, with normal data typically dominating the dataset, leading to a prevalent imbalance in favor of negative samples. Among the existing unsupervised methods in use, particularly in detecting rare anomalies, are significantly influenced by the abundance of dominant negative samples. While the above-mentioned endeavors have been continuously exploring and striving within the domain of VAD, the detection of anomalies in VAD still presents challenges with regard to two distinct variables [2]: (1) Rarity or scarcity. Anomalies are rare, making it challenging to acquire enough authentic examples for training. (2) Ambiguity or vagueness. Anomalies lack fixed semantics, as they encompass diverse events across varying subjects and scenarios. Whether an event is normal or anomalous depends on the task's context or overall scenario, thus introducing variability and uncertainty. More recently, self-supervised multi-task learning has emerged as another avenue for VAD, yielding significantly improved results [17], [18], [19], [20], [22], [23], [24]. Self-supervised learning methods aim to explore supervisory signals by learning representations from unlabeled data. It involves designing auxiliary tasks (proxy tasks) to leverage unlabeled data for self-supervised learning. These auxiliary tasks encompass various strategies, including data augmentation by manipulating parameters of existing data to transform it into new and unique samples; employing temporally aware contrastive learning to address performance issues arising from inconsistencies between individual auxiliary tasks and anomaly detection tasks [25]; combining different forms of predictive tasks or implementing various variants of a single predictive task [10], [11]; and utilizing different methods in one or more reconstruction tasks to alleviate the "identity mapping" problem in reconstruction, thereby better distinguishing anomalies [12]. The ultimate goal is to empower models to discern valuable features within the data and progressively assimilate an understanding of the inherent structure and semantic information through the execution of these tasks, leading to enhanced overall performance.

Based on the preceding discussion, Individual reconstruction- or prediction-based methods (Fig. 1(a)) struggle to capture comprehensive object representations, especially for subtle anomalies, leading to suboptimal performance. Dual-branch structures (Fig. 1(b)), which separately learn spatial and temporal features, may suffer from information loss when handling these dimensions independently. Jointly training multiple correlated tasks often outperforms single-task approaches, extracting more comprehensive features. As shown in Fig. 1(c), multiple proxy tasks can guide the model toward richer feature representations, enhancing anomaly detection performance. This synergistic approach improves the model's consistency and effectiveness in detecting subtle anomalies. In this work,



(a) Single-task (b) Double-task (c) Multi-task (ours)

Fig. 1. Illustration of difference in various task paradigms for video anomaly detection. (a) single-task paradigms lack subtle anomaly feature capture, (b) temporal-spatial separation misses feature interactions, and (c) Ours is a decoupled self-supervised multi-task learning approach.

we designed a more challenging proxy task by spatial-temporal decoupling, where we designed multiple proxy sub-tasks along the time and space dimensions to process time and spatial information at a finer granularity. First, to extract key information from the video, we employed an off-the-shelf object detector to detect each object in the video and extract the corresponding feature information. Next, for each detected object, we performed a series of operations, including cropping along the object's bounding box to construct a spatiotemporal context cube centered around the object, adapting the cube structure based on the task requirements. In the spatial dimension. ① We applied a mask operation to the segmented patches and used them as input to train the model by reconstructing the masked parts of the information. ② For each frame in the spatiotemporal context cube, we decomposed it into several equal-sized patches, scrambled the original positions of these patches, and then trained the model to predict their correct positions. These tasks help the model gain a deeper understanding of the structural aspects of spatial information and its relationship to the overall content of the video. In the temporal dimension. ③ For all frames in the spatiotemporal context cube, we shuffled their temporal order and then trained the model to predict the correct position of these frames in the original sequence. ④ We removed a specific frame from each spatiotemporal context cube and used the information from the preceding and succeeding frames to predict the removed frame. Through these operations, the model is able to capture anomalous patterns in time and further enhance its ability to understand temporal information. We believe that anomaly detection involves not only motion anomalies but also appearance anomalies. Therefore, through spatial-temporal decoupling, we perform multitask processing along the time and space dimensions to more effectively identify different types of anomalies. We highlight the main contributions as follows:

- We have decoupled the video anomaly detection task into both temporal and spatial dimensions, designing it as a multi-task learning paradigm for anomaly detection.
- We approach video anomaly detection as a multi-task learning challenge, wherein we consolidate diverse self-supervised proxy tasks from various dimensions within a unified model for joint learning.
- We were among the early adopters of decoupled self-supervised multi-task learning in the domain of video anomaly detection.

– We have conducted experiments that illustrate the superiority of our approach over state-of-the-art methods across three different traffic road scenario benchmarks.

The subsequent parts of this paper are organized in the following manner: Section II delves into the related work on self-supervised multi-task learning in video anomaly detection. Section III offers a comprehensive description of the proposed self-supervised multi-task learning method with temporal-spatial decoupling. Section IV presents a series of experiments conducted to evaluate the proposed method, along with a discussion of some of future directions and practical implications. In the end, Section V provides a summary of the methodology adopted for this paper.

II. Related Work

A. Video Anomaly Detection

The advancement of deep neural networks has significantly enhanced anomaly detection by incorporating various research paradigms, including knowledge distillation [20], [21], self-supervised learning [22], [23], [24], [25], and meta-learning [26]. These approaches collectively contribute to the improved efficiency and accuracy of capturing anomalous behaviors within complex scenes. Georgescu et al. [19] proposed an object-level approach that treats anomaly detection as a binary classification task. To overcome the issue of lacking anomaly samples and to improve performance, they created scene-independent pseudo-anomaly samples for adversarial training and binary classifier training. However, the introduction of optical flow significantly reduced the model's computational efficiency, potentially making it unsuitable for real-time processing of high frame-rate or high-resolution videos. Zhou et al. [27] proposed an appearance-motion network to improve the accuracy of video-based accident detection in congested traffic. Features are extracted through parallel appearance and motion convolutional networks, and auxiliary networks, optical flow learners, and temporal attention modules are introduced to enhance the model's ability to recognize complex traffic scenes. Additionally, a new traffic anomaly dataset is constructed and effectively validated. However, the model occasionally misidentifies non-collision events, such as vehicle malfunctions, as collisions due to the similarity in motion features. Qing et al. [24] proposed a self-supervised learning approach called hierarchical consistency (HiCo) specifically designed for untrimmed videos that contain abundant visual and semantic information. This method tailors the learning process to capture various levels of visual and semantic consistency, adapting them based on the relevance between visual and semantic components. Utilizing reconstruction or prediction has emerged as a prevalent approach within the realm of video anomaly detection. These methods are trained solely on data representing normal patterns, nevertheless, deep neural networks exhibit an impressive capacity for generalization, enabling the model to adeptly reconstruct (or predict) demanding anomaly samples as well, it introduces the challenge of over-generalization. Wu et al. [17] devised the self-supervised sparse representation (S3R) framework, exploring the synergistic effect between dictionary learning and self-supervised learning. By using dictionary learning to generate segment-level features and leveraging self-supervised learning to generate pseudo-normal/abnormal samples, the framework trains an anomaly detector. Although it effectively distinguishes normal and abnormal events in most cases, it still encounters false negatives when dealing with certain complex anomalies, such as low-frequency anomalies. Zanella et al. [28] leveraged existing vision-language models to generate textual descriptions for each test frame, along with a specific prompting mechanism to unlock the language model's capabilities in temporal aggregation and anomaly score estimation, making it an effective video anomaly detector.

B. Multi-Task Learning on VAD

As described by Zhang and Yang [29], we initially provide the definition of multi-task learning:

Definition 1 (Multi-task learning): Given m learning tasks $\{\mathcal{T}_i\}_{i=1}^{m}$ where all the tasks or a subset of them are related but not identical, multi-task learning aims to help improve the learning of a model for T_i by using the knowledge contained in the m tasks.

Multi-task models have the capacity to learn both the commonalities and differences among various tasks, resulting in improved efficiency and model quality for each individual task [30], [31]. Consequently, compared to single-task models, the risk of over fitting diminishes, while the generalization capability strengthens. Georgescu et al. [20] pioneered the treatment of video anomaly detection as a multi-task learning endeavor. By combining knowledge distillation and several self-supervised proxy tasks into a unified architecture, they effectively tackled the complex challenge of anomaly detection. Baradaran and Bergevin [30] incorporated three complementary proxy tasks, establishing two branches for future semantic segmentation and optical flow magnitude estimation to achieve anomaly detection. Additionally, they innovatively integrated factors such as the direction of object motion and distance from the camera into the scope of anomaly detection. Barbalau et al. [23] revisited the self-supervised multi-task learning framework originally proposed in [20]. Building upon this foundation, they explored alternative proxy tasks and adopted the convolutional vision Transformer (CvT) [31] as the backbone, resulting in further enhanced performance of the model. Therefore, leveraging the characteristics and advantages of multitask learning, we designed several targeted proxy tasks and trained them jointly. The model can fully explore the intrinsic correlations and complementarities between the tasks, significantly enhancing its ability to understand and represent the data. Multitask learning enables knowledge sharing across different tasks, allowing the model to simultaneously learn and extract information from multiple perspectives. This not only enhances the model's generalization performance in complex scenarios but also facilitates knowledge transfer and sharing between tasks.

C. Self-Supervised Learning on VAD

Self-supervised learning (SSL) is a general learning paradigm with the aim of automatically constructing supervision signals from raw data, without external labels or manual annotations. Its core principle is to leverage the inherent structure and information within the data for the learning process. SSL can be divided into two main methods: constructing proxy tasks and conducting contrastive learning. In proxy tasks for images, researchers design a series of tasks such as data rotation [32], jigsaw puzzles [33], prediction, or reconstruction [34], [35] to enable the model to learn visual features from the data. Unlike image proxy tasks, video proxy tasks have their unique attributes—temporal information. In video proxy tasks, the focus is often on temporal information, such as frame sequence ordering [22], frame playback speed [36], and the arrow of time [20], [37], among others. Wang et al. [22] introduced a novel proxy task for predictive sequence modeling that disentangles spatiotemporal components by shuffling spatial and temporal cubes, simplifying the self-supervised learning framework and achieves remarkable performance improvements, although it requires substantial computational resources due to the extensive use of permutation-based data. Shi et al. [38] proposed a progressive learning strategy based on the difficulty of proxy tasks, gradually introducing multiple proxy tasks in ascending order of task difficulty to improve performance. If a proxy task is more challenging, they typically encourage the model to learn more complex features; the performance might be better, which is exactly what our method achieves. In contrastive learning, models are tasked with extracting features from the same sample or distinct samples and learning by comparing the similarity of these extracted features [16]. Liu et al. [26] proposed a meta-learning method based on ensemble uncertainty for anomaly detection in road scenes, particularly suitable for autonomous driving. This method implicitly quantifies uncertainty by measuring the differences between different models, detecting out-of-distribution (OoD) samples that deviate significantly from the training data distribution. However, due to its reliance on the outputs of multiple semantic segmentation models, inference requires running multiple models, which may lead to longer inference times, posing certain limitations, especially in real-time road monitoring systems. Wang et al. [39] used an autocorrelation mechanism to map video features into sub-feature spaces, supervised by contrastive loss to detect anomalous videos. Still, it demands a substantial amount of training data and task-specific data augmentation strategies. Inspired by the significant success of self-supervised learning in the field of image representation learning, we attempt to extend this paradigm to the domain of anomaly detection, exploring its potential in video analysis. By applying self-supervised multi-task learning to video, we can fully exploit the latent information within the video, thereby better learning video representations and features. It will bring new perspectives and challenges to research and applications in the field of video anomaly detection.

III. Methodology

A. Overview

Overview. An overview of the spatiotemporal decoupled self-supervised multi-task learning network that we propose at the object level is depicted in Fig. 2. The basic parts of our approach are as follows: generating spatiotemporal context cubes, developing spatiotemporal decoupled proxy tasks, and joint training. To begin, we utilize an off-the-shelf object detector such as YOLOv3 or YOLOv5 to obtain a series of bounding boxes related to the object in the adjacent frames of the video $\mathbb{I} = \{I_{i-l}, \cdots, I_{i-1}, I_i, I_{i+1}, \cdots, I_{i+l}\}$, We then perform cropping and resizing operations along these bounding boxes for the same object to generate the spatiotemporal context cube. After that, in order to adequately take into account the attributes of the spatial and temporal dimensions, we apply an array of explores to develop adaptable proxy tasks. Within the spatial dimension, two proxy subtasks are introduced: "the spatial patches masking" and "predicting the positions of the spatial patches". The model learns information of different spatial visual features through such tasks, thereby improving its ability to pick up spatial structures and patterns in videos. Within the temporal dimension, two proxy subtasks are established: "predict missing boxes" and "predict temporal position". These tasks contribute to the model's learning of the temporal evolution and changes that occur during abnormalities. In the end, we train the aforementioned proxy subtasks jointly by a shared backbone. Diverse proxy subtasks are able to interact and connect by a shared visual representation of features. By acquiring more extensive and detailed feature information from various proxy subtasks, the model enhances its performance. Furthermore, by sharing the backbone, conflicts and competition among distinct proxy tasks are minimized, which ended in a more stable and efficient learning procedure for the model. The symbols and their corresponding descriptions are presented in Table I of this paper.

1) Backbone Architecture: Drawing on prior works [20], [22], [40], we employ 3D convolutional neural network (3D CNN) and Transformer blocks to construct the backbone network, with each proxy subtask having its corresponding prediction head. Initially, we model the spatiotemporal representation of the input using a 3D CNN block. Each 3D block consists of two $3 \times 3 \times 3$ 3D convolutional layers and a 3D max-pooling layer, with each convolutional layer followed by a batch normalization (BN) layer and a ReLU activation function. Following that, we provide Transformer blocks that are capable of effectively processing video sequences, the output produced by the Transformer blocks is subsequently subjected to global pooling in order to generate the final result.

B. Temporal-Spatial Decoupled Self-Supervised Multi-Task Learning

Spatial dimension—Proxy subtask I. Spatial patches masking. Within the field of self-supervised learning, He et al. [41] devised a straightforward yet exceptionally efficient method: they reconstruct the missing pixels of the input image after randomly masking a portion (or patch) of it. A significantly enhanced performance has been achieved through the implementation of this approach. Encouraged by this, we implemented the mask reconstruction task in the frames at the object level and optimized the network utilizing the L2 loss. In the testing phase, a random subset of the patches

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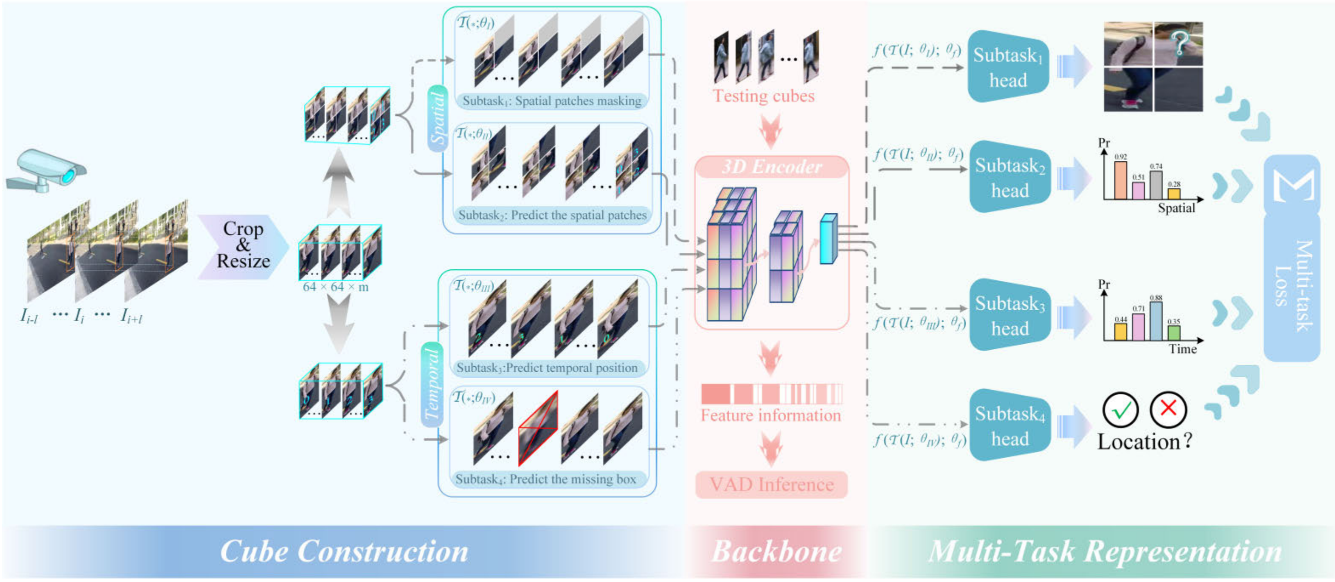

Fig. 2. Method overview. We propose an object-centric spatiotemporal decoupled self-supervised multi-task learning network, which consists of three main components: the construction of spatiotemporal context cubes; employing diverse strategies to generate flexible proxy tasks in both temporal and spatial dimensions; and joint training using a shared backbone. We can simultaneously learn temporal and spatial features within a unified framework, thus better capturing anomalies in videos.

TABLE I
NOTATIONS AND THEIR DESCRIPTIONS INVOLVED IN THE PROPOSED METHOD

Notation	Description
$\mathbb{I}$	A set of original spatiotemporal context cubes for object-level frames
$\mathcal{T}(*;\theta_I)$	The spatiotemporal context cubes transformed according to proxy subtask I
$\mathcal{T}(*;\theta_{II})$	The spatiotemporal context cubes transformed according to proxy subtask II
$\mathcal{T}(*;\theta_{III})$	The spatiotemporal context cubes transformed according to proxy subtask III
$\mathcal{T}(*;\theta_{IV})$	The spatiotemporal context cubes transformed according to proxy subtask IV
$\hat{I}^I$	The output generated by proxy subtask I
$\hat{I}^{II}$	The output generated by proxy subtask II
$\hat{I}^{III}$	The output generated by proxy subtask III
$\hat{I}^{IV}$	The output generated by proxy subtask IV
$f(*;\theta_f)$	The backbone of our model, the 3D encoder
$\mathcal{L}_{sub*}$	The optimization functions associated with distinct proxy subtasks
$Score_{st*}$	The object-level anomaly scores corresponding to different proxy subtasks
$\mathcal{L}_{total}$	The joint optimization function for the whole network
$Score(I)$	The object-level anomaly score for the overall task
$Score^{fra}$	The frame-level anomaly score

in the input (64 × 64 in our settings) is masked for reconstruction. Finally, the anomaly score, which is calculated using mean absolute difference (MAD), is utilized for determining whether the current frame is anomalous by calculating the difference between the original image and the completed reconstruction.

$$\hat{I}^I = f\left(\mathcal{T}\left(I;\theta_I\right);\theta_f\right) \tag{1}$$

$$\mathcal{L}_{sub1} = \left\| I - \hat{I}^I \right\|_2^2 \tag{2}$$

$$Score_{st1} = \frac{1}{m}\sum\nolimits_{i=1}^{m}\left| I_i - \hat{I}_i^I \right| \tag{3}$$

where, the ground truth of the i-th object-level frame is denoted as I_i, $\hat{I}^I$ represents the output obtained from the proxy subtask I, $\mathcal{T}\left(*;\theta_I\right)$ represents the spatiotemporal context cube with masked patches, $f\left(*;\theta_f\right)$ represents the backbone of our model, the 3D encoder, and the number of frames in the cube is denoted by m.

Proxy subtask II—Predict the positions of the spatial patches. First, each frame within the spatiotemporal context cube is split into n × n the same patches, which are subsequently shuffled. To ensure that the content and relative position of each patch within the context cube are correct, we set the absolute positions of the patches to be arranged sequentially from left to right and from top to bottom. Concurrently, so as to preserve the natural temporal order, the relative positions of each frame along the temporal axis of the spatiotemporal context cube remain constant. After inputting it into the backbone for training, use the spatial patches position classifier to predict the absolute position of each patch in every frame, and optimize the network using cross-entropy loss, as shown in equation (5). During the testing phase, we use context cubes containing the correct patch position sequences. The model predicts the position of patches in each frame and generates a position matrix. In the position matrix, the diagonal values should be the largest in their respective columns, indicating the probability that the corresponding patch is in the correct position. Since the model is trained only with normal data, when encountering anomalous events during testing, the model often produces large prediction errors at the correct anomalous locations, resulting in significantly lower confidence in the predicted outcome at these locations, which leads to the detection of anomalies at those positions. Therefore, for anomaly scores, the predicted probabilities along the diagonal of the matrix are used as the anomaly scores for these patches. The matrix predicted by the spatial patches classifier is denoted as Z_s. The diagonal of the matrix is extracted using diag(·), and the minimum value is used to obtain $Score_{st2}$. This value denotes the object-level anomaly score. In a nutshell, the object-level anomaly score is determined by selecting the minimum score along the diagonal of the matrix. This is due to the fact that a sample may be considered abnormal if a frame or patch is incorrectly predicted.

$$\hat{I}^{II} = f\left(\mathcal{T}\left(I;\theta_{II}\right);\theta_f\right) \tag{4}$$

$$\mathcal{L}_{sub2} = \frac{1}{n^2}\sum\nolimits_{s=1}^{n^2} CE(I_s, \hat{I}_s^{II}) \tag{5}$$

$$Score_{st2} = min\left(diag\left(Z_s\right)\right) \tag{6}$$

where, $\hat{I}^{II}$ represents the output obtained from the proxy subtask II, $\mathcal{T}(*;\theta_{II})$ represents the spatiotemporal context cube with spatially shuffled patches. To align with the overall trend of anomaly scores for the entire task, we invert the frame-level anomaly scores. It implies that when the overall anomaly score takes on higher values, it can be considered as an anomaly, and vice versa.

Temporal dimension—Proxy subtask III. Predict temporal position. Stephen Hawking [42], in his seminal work "A Brief History of Time" introduced the concept of the "arrow of time," which suggests the reality of a distinct trajectory for time and provides substantiation for the inexorable progression of forward-moving time that is unreversible. Influenced by it, we adopted a strategy to address the spatiotemporal context. To be more precise, we shuffle and reorder the frames within the spatiotemporal context cube along the timeline and then input them into the model for training. By predicting the correct position of each frame, the model is guided to more comprehensively capture the temporal relationships between frames. During testing, we input context cubes with the correct temporal order to generate the temporal position prediction matrix Z_t. Each element in the matrix represents the predicted position of the corresponding frame on the time axis. Afterward, we use the diagonal prediction probability of the prediction matrix as the anomaly score. In terms of network optimization, we utilize a cross-entropy loss function as follows:

$$\hat{I}^{III} = f\left(\mathcal{T}\left(I;\theta_{III}\right);\theta_f\right) \tag{7}$$

$$\mathcal{L}_{sub3} = \frac{1}{m}\sum\nolimits_{i=1}^{m} CE(I_i, \hat{I}_i^{III}) \tag{8}$$

$$Score_{st3} = min\left(diag\left(Z_t\right)\right) \tag{9}$$

where, $\hat{I}^{III}$ represents the output obtained from the proxy subtask III, $\mathcal{T}(*;\theta_{III})$ represents the spatiotemporal context cube with shuffled frames.

Proxy subtask IV. Predict the missing box. Discarding (or removing) a frame from the spatiotemporal context cubes and predicting the frame using its contextual information. This includes predicting not only the next frame based on the preceding video sequence but also an intermediate frame using the frames before and after it, and predicting the content of a prior video frame using subsequent frames. This design guarantees a comprehensive prediction process by including various perspectives to capture the features of the object frame. As a result of object-level frame prediction, the prediction process is less susceptible to background interference, which improves the accuracy of the missing frame prediction. During training, we optimize the network using the L2 loss function, and we utilize mean squared error (MSE) as the anomaly score ($Score_{st4}$) to determine anomalies, as shown below:

$$\hat{I}^{IV} = f\left(\mathcal{T}\left(I;\theta_{IV}\right);\theta_f\right) \tag{10}$$

$$\mathcal{L}_{sub4} = \left\| I - \hat{I}^{IV} \right\|_2 \tag{11}$$

where, $\hat{I}^{IV}$ represents the output obtained from the proxy subtask IV, $\mathcal{T}(*;\theta_{IV})$ represents the spatiotemporal context cube with missing frames.

C. Joint Loss Function

To train the network, we utilize a 3D backbone with Transformer for joint optimization on all four of the aforementioned proxy subtasks. The loss function therefore associated with joint training is as follows:

$$\mathcal{L}_{total} = w_1\mathcal{L}_{sub1} + w_2\mathcal{L}_{sub2} + w_3\mathcal{L}_{sub3} + w_4\mathcal{L}_{sub4} \tag{12}$$

where, $w_{1,\ldots,4}$ represents the weights during joint training, determining the trade-off among the four proxy tasks. The training pseudo code for the method we employ is provided by Algorithm 1.

D. VAD Inference

When testing, we utilize an object detector to perform object detection on each frame. For each object, a series of operations, including cropping, are conducted to generate the corresponding object-level spatiotemporal context cube. The model is trained using each spatiotemporal cube as input, and its outputs correspond to the following subtasks: $Score_{st1}$, $Score_{st2}$, $Score_{st3}$, and $Score_{st4}$. In the spatial dimension, for the spatial patch masking subtask, we use MAD as the anomaly score. In the predict positions of the spatial patches subtask, we choose the probabilities along the diagonal of the spatial prediction matrix as the anomaly score. In the temporal dimension, for the predict temporal position subtask, we choose the probabilities along the diagonal of the temporal prediction matrix as the anomaly score. In the missing frame prediction subtask, we still use MSE as the anomaly score to determine anomalies. The final anomaly score of an object is calculated by averaging the anomaly scores supplied by each prediction head:

$$Score(I) = \frac{1}{4}\left(Score_{st1} + Score_{\overline{st2}} + Score_{\overline{st3}} + Score_{st4}\right) \tag{13}$$

where, $Score_{\overline{st2}}$ and $Score_{\overline{st3}}$ represent negation, and higher values indicate anomalies. Because the initial detection is

Algorithm 1 Training pseudo code of temporal-spatial decoupled self-supervised multi-task learning

Input: $\mathcal{D}$ the training dataset with only normal frames; object-centric spatiotemporal context cubes; Cube size; the learning rate; 3D Encoder.
Output: optimally trained model.
Initialize all parameters.
for the number of training epochs **do**
  Sample object-centric spatiotemporal context cubes from $\mathcal{D}$.
  Function multi − task$(\cdot)$:
    # Proxy subtask I of spatial patches masking.
    $\mathcal{L}_{\text{sub1}} \leftarrow \left\| \mathrm{I} - \hat{I}^{\mathrm{I}} \right\|_2^2$
    # Proxy subtask II of predicts the positions of the spatial patches.
    $\mathcal{L}_{\text{sub2}} \leftarrow \mathrm{CE}(\mathrm{I_s}, \hat{I}_s^{\mathrm{II}})$
    # Proxy subtask III of predicts temporal position.
    $\mathcal{L}_{\text{sub3}} \leftarrow \mathrm{CE}(\mathrm{I_i}, \hat{I}_i^{\mathrm{III}})$
    # Proxy subtask IV of Predict the missing box.
    $\mathcal{L}_{\text{sub4}} \leftarrow \left\| \mathrm{I} - \hat{I}^{\mathrm{IV}} \right\|_2$
  return $w_1\mathcal{L}_{\text{sub1}} + w_2\mathcal{L}_{\text{sub2}} + w_3\mathcal{L}_{\text{sub3}} + w_4\mathcal{L}_{\text{sub4}}$
  $\mathcal{L}_{\text{total}} \leftarrow$ multi-task$(\cdot)$
  # parameter optimization
  Update network parameters using the optimizer Adam.
end for

at the object level, it's easy to locate the anomaly region. As there are multiple objects in the video frame, each with its own anomaly score, the overall anomaly score for the entire video frame is determined by the maximum score among the corresponding anomaly maps.

Let the frame-level anomaly score be $\mathrm{Score}^{\mathrm{fra}}$. For each video frame where j objects are detected, the overall anomaly score is as follows:

$$Score^{fra} = \begin{cases} max\left\{Score(I)_1, Score(I)_2, \ldots, Score(I)_j\right\}, \\ \qquad j > 0 \\ 0, \qquad j = 0 \end{cases} \tag{14}$$

If there are no objects detected in a frame, the anomaly score is assigned to 0.

IV. Experiments

A. Datasets

In this section, we evaluate our proposed method on three publicly available datasets: the CUHK Avenue dataset [43], the UCSD Pedestrian dataset [44], and the ShanghaiTech Campus dataset [45]. We conduct both qualitative and quantitative analyses of the experimental results to demonstrate the effectiveness of our proposed approach.

- The UCSD Pedestrian dataset [44] consists of two subsets: UCSD Ped1 and UCSD Ped2, each corresponding to a specific scenario. UCSD Ped1 includes 34 training clips and 36 testing clips, while UCSD Ped2 has 16 training clips and 12 testing clips. Both subsets provide frame-level and pixel-level ground truth annotations. In this dataset, any non-pedestrian objects or abnormal movements of pedestrians are considered anomalies.

TABLE II
Comparison Anomaly Detection Datasets. (#Frames\Videos)

Dataset	Scenes No.	Training No.	Testing No.	Resolution	Abnormal events No.
UCSD Ped1 [44]	1	6,800\34	7,200\36	158×238	40
UCSD Ped2 [44]	1	2,550\16	2,010\12	240×360	12
Avenue [43]	1	15,328\16	15,324\21	360×640	47
ShanghaiTech [45]	13	274,515\330	42,883\107	480×856	130

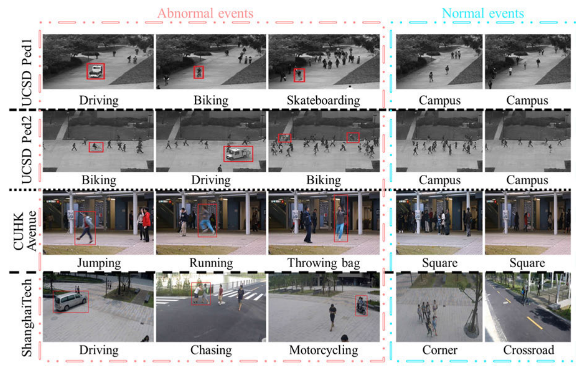

Fig. 3. Examples of both anomalous and normal events in each dataset.

- CUHK Avenue [43] is released by the Chinese University of Hong Kong (CUHK) in 2013. It consists of 16 training videos and 21 testing videos, with a resolution of 640 × 360 pixels. The video clips cover various indoor and outdoor scenes and contain 47 different types of anomalies, including running, crowding, loitering, and more.
- The ShanghaiTech Campus dataset [45] provides a variety of scenarios and anomalous behaviors. It contains video clips of both normal and abnormal behaviors, which includes traffic accidents, sudden occurrences, assemblies, and more. The dataset consists of 330 training videos and 107 testing videos, with 13 diverse scenes and a wide range of abnormal categories. The resolution of the videos is 856 × 480 pixels. The dataset contains video clips with various lengths, perspectives, conditions of lighting, and other details, making them more representative of the complexity of the actual world.

Fig. 3 depicts normal and anomalous samples from the aforementioned datasets. The illustrations that were in the figure depict video snippets of various scenarios, including normal and abnormal actions such as walking, congestion, battling, and traffic accidents, etc. Table II displays some statistics regarding the various datasets, including the number of video segments, the number of frames, and the kinds of abnormal behaviors present in each dataset. This information provides an overview of the characteristics and range of the datasets utilized in the evaluation.

B. Evaluation Metrics and Implementation Details

As per the evaluation metric that is commonly accepted and used within the VAD community [20], [34], In video anomaly detection, area under the curve (AUC) is a widely employed evaluation metric. It assesses a model's ability to distinguish between positive and negative samples by measuring the area under the receiver operating characteristic (ROC) curve, depicted from the starting point (0,0) to the endpoint (1,1). By employing AUC, subjective assumptions can be circumvented during the threshold selection process. AUC comprehensively considers all possible thresholds, providing a more holistic evaluation of model performance compared to other methods using fixed thresholds (e.g., accuracy and recall), especially when dealing with imbalanced datasets. Therefore, we employ AUC to evaluate our proposed method.

1) Implementation Details: In order to guarantee fairness and generality in comparison, we followed the setup of previous outstanding works [22], [23], [46] and used YOLOv3 as the object of interest extractor. For different datasets, we set different confidence thresholds according to their characteristics to filter out low-confidence objects. Specifically, for the medium-sized dataset Avenue and the large-scale dataset ShanghaiTech, we set the confidence threshold at 0.8 to maintain high accuracy. For the smaller and earlier-released dataset, UCSD Ped2, due to its lower pixel resolution and relatively smaller data volume, we set the confidence threshold at 0.5. We resized the input data to 64×64 and set the batch size to 128. The Adam optimizer is employed with a learning rate of 2e-4. The hyperparameters $\beta 1 = 0.9$ and $\beta 2 = 0.999$ are set up for the Adam optimizer, with all other parameters remaining at their default settings. To ensure that each task contributes sufficiently during the multi-task learning process, we assign equal weights to the hyperparameters $w_{1,\dots,4}$ in Equation (12). The entire network training and testing are conducted end-to-end on an NVIDIA GeForce RTX 3090 GPU.

C. Comparative Evaluation With State-of-the-Art (SOTA) Methods

The comparison between our proposed method and the existing state-of-the-art methods on three benchmark datasets (Avenue, ShanghaiTech, and UCSD Ped2) is represented in Table III. These methods are categorized into single-proxy tasks (based on reconstruction/prediction), hybrid-proxy tasks, and multi-proxy tasks according to different paradigms. We found that whether it is a single proxy task based on reconstruction or prediction, or a hybrid proxy task combining the both, the performance generally lags behind that of self-supervised multi-proxy tasks, and this gap becomes increasingly pronounced as the scenes become more complex. In the self-supervised multi-task paradigm, the aggregation of multiple complex task features enables the method to fully exploit the data itself as supervisory information, thereby demonstrating superior performance on larger-scale and more complex datasets. Results on Ped2: the performance of various proxy task methods is consistently above 90%. Compared to the previous state-of-the-art method [57], our method offers highly competitive performance, with a mere 0.1% disparity. Due to the relatively old release date of the Ped2 dataset and its lower pixel resolution, the images are not particularly clear. Therefore, when performance approaches a bottleneck, the performance differences between methods are not so pronounced.

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS. THE PERFORMANCES OF THE COMPARED METHODS ARE SOURCED FROM THEIR ORIGINAL PAPERS

Paradigm	**Method**	**Datasets (AUC %)**		
		Ped2	**Avenue**	**SH.T**
Reconstruction-based	Conv-AE[47]	90.0	70.2	–
	StackRNN[45]	92.2	81.7	68.0
	Mem-AE[5]	94.1	83.3	71.2
	MNAD-Rec[35]	90.2	82.8	69.8
	VEC[46]	97.3	89.6	74.8
	DGN[48]	95.7	86.8	73.0
	GAE[49]	97.6	89.9	74.7
	STATE[9]	–	90.3	77.8
	MAAM[7]	97.7	90.9	71.3
Prediction-based	Frame-Pred[34]	95.4	85.1	72.8
	MNAD-Pred[35]	97.0	88.5	70.5
	CT-D2GAN[50]	97.2	85.9	77.7
	DLAN-AC[51]	97.6	89.9	74.7
	STCEN[52]	96.9	86.6	73.8
	STM[10]	99.4	90.5	94.3
	USTN-DSC[11]	98.1	89.9	73.8
	MPED-RNN[53]	–	–	73.4
Hybrid-based	sRNN[54]	92.2	83.5	69.6
	HF^2-VAD[55]	99.3	91.1	76.2
	DEDDnet[56]	98.1	89.0	74.5
	DMAD[57]	99.7	92.8	78.8
	AMSRC[58]	99.5	93.8	76.6
SSL	CAC[39]	–	87.0	79.3
	SSMTL[20]	97.5	91.5	82.4
	Jigsaw-Puzzle[22]	99.0	92.2	84.3
	AI-VAD[59]	99.1	93.3	85.9
	MoPRL[60]	–	–	83.4
	HSC[61]	98.1	93.7	83.4
	SLMPT[38]	97.6	90.9	78.8
	Ours	**99.6**	**95.9**	**87.7**

This, in turn, illustrates the effectiveness and superiority of our method. Results on Avenue: consistent advancements in VAD and the ensuing competition among different approaches have undeniably resulted in substantial performance enhancements, enabling methods to rival one another and go close to or even surpass the 90% threshold [9], [11], [20], [46], [57], [59]. Nevertheless, in comparison to these approaches, our method's performance still reaches the current optimal level, specifically 95.9%, compared to the previous state-of-the-art self-supervised method [61], our performance surpasses it by around 2%. This indicates the benefits and potential shown by the synergy between multiple tasks and the logical combination of proxy tasks in the self-supervised learning setting. Results on ShanghaiTech: In terms of scene complexity and data scale, the ShanghaiTech dataset is more intricate and expansive in comparison to the ped2 and Avenue. However, we unveiled the most optimal result at 87.7%, which is an almost 2% improvement over the previous state-of-the-art approach [59]. This also demonstrates the superiority of self-supervised learning, which decouples the spatiotemporal dimensions of VAD and integrates expression features from multiple perspectives to optimize the model's performance.

TABLE IV

COMPARISON WITH THE OPTIMAL METHODS USING DIFFERENT MODALITIES AS INPUTS. WHERE, “i” DENOTES INPUT AS RGB IMAGE, “o” REPRESENTS INPUT AS OPTICAL FLOW, “s” SIGNIFIES INPUT AS SEMANTIC MAP, AND “g” DENOTES INPUT AS GRADIENT MAP. “d” REPRESENTS DEPTH MAP

Method	Modality	Datasets (AUC %)		
		Ped2	Avenue	SH.T
AMC[62]	$i + o$	96.2	86.9	–
object-centric[18]	$i + o$	94.3	87.4	78.7
HF^2-VAD[55]	$i + o$	99.3	91.1	76.2
STCEN[52]	$i + o$	96.9	86.6	73.8
BDPN[63]	$i + o$	98.3	90.3	78.1
AMSRC[58]	$i + o$	99.5	93.8	76.6
FVPSF[64]	$s + o$	97.8	–	85.3
STPT[65]	$i + o$	98.9	92.6	77.1
VCC[66]	$i + g$	97.3	89.6	74.8
VCC[66]	$i + o$	99.0	92.2	80.2
MTL[30]	$i + o + d$	97.8	–	89.1
Ours	i	**99.6**	**95.9**	**87.7**

In Table IV, we compared our method with state-of-the-art approaches that use different modalities as inputs. From the table, we can observe that despite prior methods using a single modality (such as optical flow/RGB image) or combinations of different modalities (such as RGB image + optical flow /semantic segmentation map, etc.) as inputs, our method remains competitive. Although the work [64] employed the combination of optical flow and semantic segmentation maps to reduce background interference in input data, making the model more focused on dynamically changing foreground objects, our approach continues to demonstrate superior performance. While the performance of [30] surpasses ours on ShanghaiTech, it’s essential to note that they employed three different modalities (RGB image + optical flow + depth map), combining them, whereas our method solely utilizes RGB image. The utilization of multiple modalities not only enhances the capture of object appearance and motion but also improves the performance of the model to a significant extent. Inputs from modalities other than images reduce background interference, benefiting the performance improvement of the model, this contributes valuable insights to our future work. The aforementioned analysis suggests that multi-task learning can indeed acquire richer feature representations than a single task, and even outperforms features learned when different modalities with less noise are fused as inputs.

D. Ablation Study

In Table V, we provide a detailed presentation of the individual performance of each proxy subtask on the Avenue and Ped2 datasets. Encouragingly, each task exhibits unique advantages and achieves satisfactory results within its respective domain. This indicates that each proxy subtask in our multi-task approach contributes valuable information and features to the model training, thereby aiding in the overall enhancement of model performance. After a more detailed analysis of the data in the table, we can observe that different proxy tasks exhibit unique performance characteristics in various scenarios. The performance of the same task varies across different scenarios as well. However, these proxy tasks effectively compensate for potential limitations of the overall model under a single task by enriching feature information density from their respective perspectives. This indicates that introducing diverse proxy tasks to enrich feature information can provide more comprehensive support for VAD, ultimately leading to a significant improvement in the overall model performance.

TABLE V

PERFORMANCE ON AVENUE AND PED2 WHILE VARYING THE PROXY SUBTASKS

Subtasks	Performance (AUC %)	
	Ped2	Avenue
T_{sub1}	94.0	86.6
T_{sub2}	94.9	88.1
T_{sub3}	96.1	81.2
T_{sub4}	97.6	89.7

TABLE VI

PERFORMANCE ON AVENUE WHILE CHANGING THE PROXY SUBTASKS

Subtasks	Performance (AUC %)
$T_{sub1+ sub2}$	91.7
$T_{sub2+ sub3}$	93.4
$T_{sub1+sub2+ sub3}$	94.2
$T_{sub1+ sub2+sub3+ sub4}$	95.9

TABLE VII

COMPARISON OF EXECUTION EFFICIENCY WITH OTHER SOTA METHODS

Method	ShanghaiTech @ AUC(%)	Efficiency(FPS)
MNAD[35]	70.5	56
Mem-AE[5]	71.2	42
HF^2-VAD[55]	76.2	10
object-centric[18]	78.7	18
SLMPT[38]	78.8	72
SSMTL[20]	82.4	50
MoPRL[60]	83.4	108
SSMTL++[23]	83.8	19
MSVAD[67]	87.5	42
Ours	87.7	40

TABLE VIII

ANALYSIS OF THE IMPACT OF HYPERPARAMETER CONFIGURATIONS

Learning rate	Batch size	Ped2 @ AUC(%)
1e-3	128	98.7
2e-3		98.4
1e-4		99.2
2e-4		99.6
2e-4	64	98.9
	128	99.6
	256	99.4

1) Relevance Between Proxy Subtasks: Caruana [68] once made such a definition: If task A as auxiliary task can improve the performance of the model in task B, then the task A and task B are associated, that is, “related”. The performance variations resulting from various subtask combinations on Avenue

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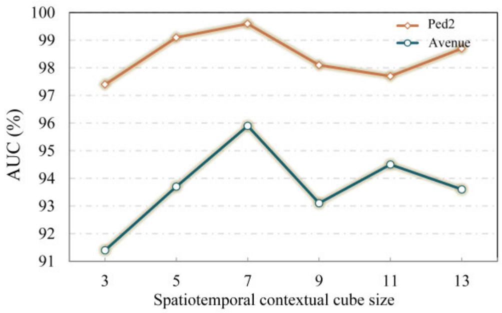

Fig. 4. Performance corresponding to different cube sizes.

are displayed in Table VI. We conducted a series of experiments initially, combining spatial dimension subtasks 1 and 2. The results indicate a significant performance improvement compared to their respective individual subtasks. Subsequently, a substantial enhancement in performance was also observed in the combined experiment that includes subtasks 2 and 3. Further, we conducted experiments involving three tasks and the combination of all tasks, the results demonstrated a gradual improvement in performance. This also indicates that the performance of anomaly detection can be substantially improved through the synergy and combination of tasks. This progressive performance improvement strongly supports the effectiveness of our approach and highlights the advantages of multi-task learning, as they can capture the complementarity and correlation between different tasks.

2) Generating Object-Centric Spatiotemporal Context Cubes: We use the bounding box of object o in the i-th frame as the center and apply the same bounding box before and after, forming a spatiotemporal context cube centered around the object. Fig. 4 presents an analysis of the impact of the quantity of frames included within the cube on the performance of the model. The frame number was configured within the range of 3 to 13, and its performance variability was showcased using the Ped2 and Avenue datasets. It can be observed that there is a beneficial correlation between the number of frames within a specific range and the performance of the model. Yet, when the quantity of frames within the cube continually increases, the performance suffers a decrease. It can be conjectured that if a context cube contains too few frames, the same object may appear in multiple cubes. This redundancy in object representation can lead to duplicate feature information, which may overwhelm the model with extraneous data during training and ultimately reduce detection accuracy. On the other hand, if the number of frames is too many, the context cube may contain multiple objects. Given the potential for complex spatiotemporal interactions between these objects, the model may encounter greater uncertainty when distinguishing between abnormal and normal behaviors, which could negatively affect the final detection. Therefore, setting a reasonable number of frames within the context cube is crucial to ensuring that the model captures sufficient object features while avoiding excessive complexity and redundant information.

3) Execution Efficiency Comparison: We compared the efficiency of our method with state-of-the-art (SOTA) methods from different paradigms, as displayed in Table VII. The results show that some methods trade higher performance for lower efficiency, primarily constrained by complex operations such as optical flow calculations, as in the SSMTL++ [23] and object-centric [18] methods. In contrast, in self-supervised multi-task methods, the simplified design of proxy tasks significantly improves runtime efficiency while maintaining relatively high performance, as demonstrated by the SLMPT [38]. Our method, within the self-supervised multi-task framework, is able to maintain competitive efficiency while ensuring high performance. The focus of our research is primarily on improving performance, and with further optimization of execution time in future work, its efficiency is expected to improve further.

4) Analysis of Hyperparameter Effects on the Model: We tuned and selected different hyperparameter configurations on the Ped2 dataset to determine the optimal model settings, as shown in Table VIII. Experimental results indicate that with a fixed learning rate, model performance fluctuates with changes in batch size. Similarly, when the batch size is held constant, adjustments to the learning rate have varying impacts on model performance, with smaller learning rates outperforming larger ones. This may be because larger learning rates tend to cause overly large updates to the model parameters, thereby affecting convergence stability. Overall, both the learning rate and batch size influence model performance to different extents, and appropriate parameter configurations can lead to improved performance.

E. Visualization and Analysis

Figs. 5, 6, and 7 give qualitative illustrations of our approach, encompassing anomaly score curves and exemplar frames depicting both normal and abnormal events. Note that we recognize and categorize normal and anomalous events in each traffic road scene, with a focus on highlighting abnormal ones. In general, when anomalous events take place, there is a noticeable rise in the curve representing the anomaly score, which then maintains a heightened level throughout the duration of the abnormality. Once the event or behavior that gives rise to the anomaly ceases, the curve rapidly descends to a decreased level and maintains a relatively stable trend. According to them, the method we have used has great sensitivity to anomalies and can efficiently detect anomalous events.

Fig. 5 illustrates how anomaly scores change during normal and abnormal events in the UCSD Ped2 dataset. When the video exclusively records pedestrians, the anomaly score curve continually holds a lower level, however, it is seen that the anomaly score experiences a significant spike as soon as a rider enters the frame, and this heightened level is sustained until the rider exits the frame. Fig. 6 illustrates how anomaly scores change during normal and abnormal events in the CUHK Avenue dataset. In Fig. 6, normal frames depict regular pedestrian activities. When sudden abnormal behaviors occur (e.g., a child running back and forth or a person pushing a bicycle), the anomaly score curve sharply rises, whereas when the subject of the abnormal behavior exits the frame (the child runs out of it), the curve rapidly drops. In a similar vein,

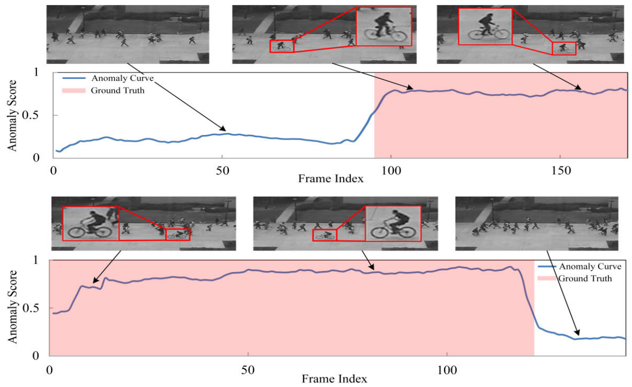

Fig. 5. Anomaly score curves and anomaly localization examples by our method on UCSD Ped2. The light red-shaded region and the white area correspondingly represent the ground truth of anomalies and normal events.

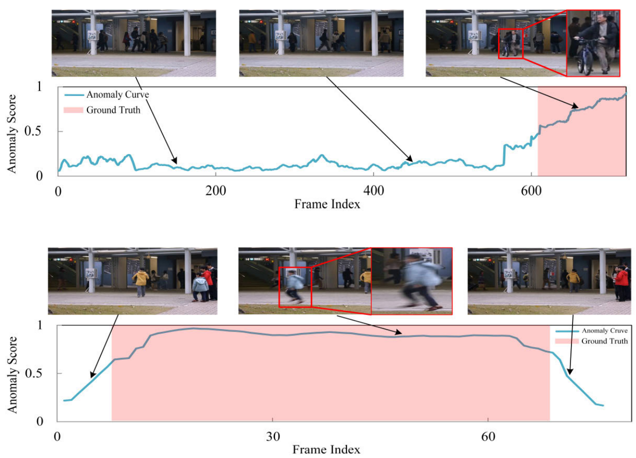

Fig. 6. Anomaly score curves and anomaly localization examples by our method on CUHK Avenue. The light red-shaded region and the white area correspondingly represent the ground truth of anomalies and normal events.

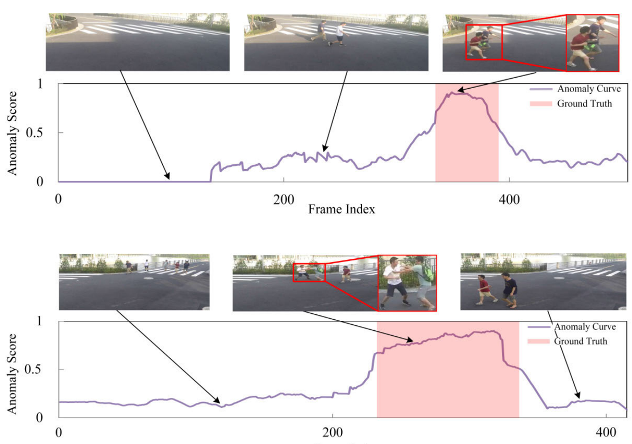

Fig. 7. Anomaly score curves and anomaly localization examples by our method on ShanghaiTech. The light red-shaded region and the white area correspondingly represent the ground truth of anomalies and normal events.

Fig. 7 illustrates the variation of anomaly scores for normal and abnormal events on the ShanghaiTech dataset. In normal frames, pedestrians walk normally at the intersection without any abnormal behavior, and the anomaly score curve keeps at a low level. However, in the case of a sudden abnormality, such as a pedestrian's bag being snatched or a physical altercation at an intersection, the score curve suffers a rapid and significant rise. After the entire anomalous event is over, the curve promptly reverts back to a lower and stable state.

F. Discussion

In this section, we will delve into the significant advantages brought by our proposed method, analyze its feasibility in practical applications, while also briefly introducing other possible methods in the field of video anomaly detection. Through comparative analysis, we aim to comprehensively evaluate the effectiveness of existing methods, while identifying potential areas for improvement and research gaps.

We perform spatial-temporal decoupling of the video anomaly detection task to separately capture temporal and spatial information in the time and space dimensions. In the temporal dimension, the focus is on the temporal sequence patterns of events, identifying dynamic changes in abnormal behaviors. In the spatial dimension, attention is given to the spatial features and positional changes of objects, revealing spatial differences in abnormal events. By constructing multiple proxy tasks related to each dimension, the model can learn and process features at different levels, enhancing its ability to tackle complex anomaly detection tasks. This approach strengthens spatial-temporal feature understanding through joint learning, improving robustness, generalization, and efficiency. In practical applications, the spatial-temporal decoupling method is highly feasible. By decoupling the time and space dimensions, it effectively captures the spatial-temporal information in videos, improving the model's ability to perform fine-grained detection of complex abnormal behaviors. This method is particularly suitable for video data such as surveillance footage and event logs, which exhibit significant temporal and spatial feature differences. Additionally, constructing multiple self-supervised proxy tasks related to time and space can promote knowledge sharing between tasks, enhance learning efficiency and generalization ability, reduce dependence on labeled data, and improve the model's adaptability in fields such as security surveillance, intelligent transportation system, and traffic anomaly early warning.

It is often said that, "stones from other hills may serve to polish jade." the same holds true for the field of VAD. With the development of technologies such as domain adaptation [69], transfer learning [70], and federated learning [71], new solutions have been proposed for multiple bottleneck problems in video anomaly detection, especially in the face of challenges like scene variations and data distribution discrepancies. These approaches provide new ideas and insights. The research on domain adaptation technology in video anomaly detection primarily focuses on enhancing the model's generalization ability to new environments and reducing the distribution gap between the source and target domains. This is particularly important when annotated data in the target domain is scarce, as it provides effective strategies for transfer and adaptation. Meta-learning methods, applied in domain adaptation and multi-task learning, have significantly improved the model's ability to quickly adapt to new tasks and domains. Transfer learning, by leveraging knowledge from one task, enhances the model's generalization capability in other tasks. This is especially useful in scenarios with limited labeled data, effectively addressing the data scarcity issue. Moreover, by using methods such as similarity analysis to assess the relationships between tasks, we can optimize model selection in transfer learning, thereby improving detection performance. This is particularly evident in anomaly detection tasks, where leveraging knowledge from traffic surveillance scenarios can significantly enhance model performance.

From the brief discussion above, it is evident that there are still many directions in the field of video anomaly detection that are worth exploring and researching in depth. These methods not only provide valuable insights for addressing the unique challenges in video anomaly detection but also effectively enhance the performance of the model in several aspects.

V. Conclusion

In the field of intelligent transportation, traffic anomaly detection plays a critical role in enhancing road safety and optimizing traffic management efficiency. In this paper, we propose an object-level spatiotemporal decoupled self-supervised multi-task learning network for video anomaly detection. Specifically, we decouple the task into spatiotemporal multi-task learning by constructing different forms of spatiotemporal context cubes in both temporal and spatial dimensions for joint learning of multiple tasks. Through experimental validation and ablation analysis, we comprehensively evaluate the effectiveness of the proposed method. The experimental results indicate that our approach attains optimal performance on widely utilized benchmark datasets, regardless of whether it involves reconstruction, prediction, or other hybrid methods. Our methodology maintains a high level of competitiveness, even when compared to multimodal input approaches. Furthermore, a succession of targeted ablation experiments confirmed the relevance and efficacy of the different components of our method. Although our method demonstrates outstanding capabilities, it is not devoid of boundaries. When dealing with subtle or more covert anomalies, relying on object detectors may have certain limitations. Therefore, future research should focus on improving the model's ability to detect these subtle anomalies while also optimizing the relevance of task agents and the interaction representations between tasks to further enhance the potential application of video anomaly detection in intelligent transportation surveillance systems, ultimately, providing a more robust technical foundation for intelligent traffic monitoring and safety management.

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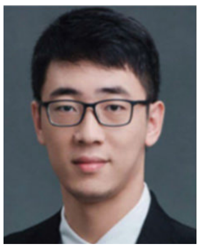

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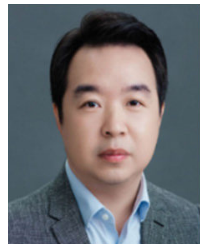

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